NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

NTU Academy for Professional and Continuing Education

AI Adoption & Workforce Impact Data Pipeline

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

AI Adoption & Workforce Impact

Data Pipeline Project — Executive Presentation

Agenda

- Executive Summary
- Business Value Proposition
- Technical Solution Overview
  - Automated ELT: Technology & Workflow
  - Star Schema Design & Structure
  - Data Quality Testing: dbt + Custom SQL + Great Expectations
  - Global AI Adoption & Workforce Impact Key Findings & Business Insights
- Risk and Mitigation
- Q&A Handling

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Executive Summary

The Challenge

- Organizations globally are investing heavily in AI, yet most lack the analytical infrastructure to measure ROI, benchmark performance, or make evidence-based adoption decisions

Which companies benefit most from AI, and how?

The Solution: End-to-End Strategic Intelligence

- An end-to-end analytics pipeline translating raw global data into actionable board-level intelligence.

DATASET OVERVIEW

	value
total_records	150000
unique_companies	10000
unique_time_periods	16
avg_productivity_change	9.27
avg_revenue_growth	4.61
avg_ai_roi_index	1.14
total_jobs_displaced	16705476
total_jobs_created	17682376
total_net_jobs_change	976900

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Key Strategic Outcomes

Maturity Drives Returns: Companies in the Scaling and Transforming stages achieve 2–3x higher AI ROI compared to early-stage adopters.

Industry Leaders: While Technology leads in productivity, traditional sectors like Logistics, Manufacturing, and Retail show the highest AI ROI index.

Responsible AI Deployment: Establishing AI ethics governance reduces job displacement rates by 18%, protecting talent while enabling automation

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Business Value Proposition

1. **Evidence-Based Decision Making**
   - Replace "gut-feel" investments with data-driven insights.
2. **Performance Benchmarking**
   - Measure your organization against cross-industry and regional benchmarks to understand your position on the adoption curve.
3. **Productivity-First Growth**
   - Prioritize AI Maturity over quantity of projects.
4. **Optimized Workforce Strategy**
   - Quantify automation’s impact on your people.

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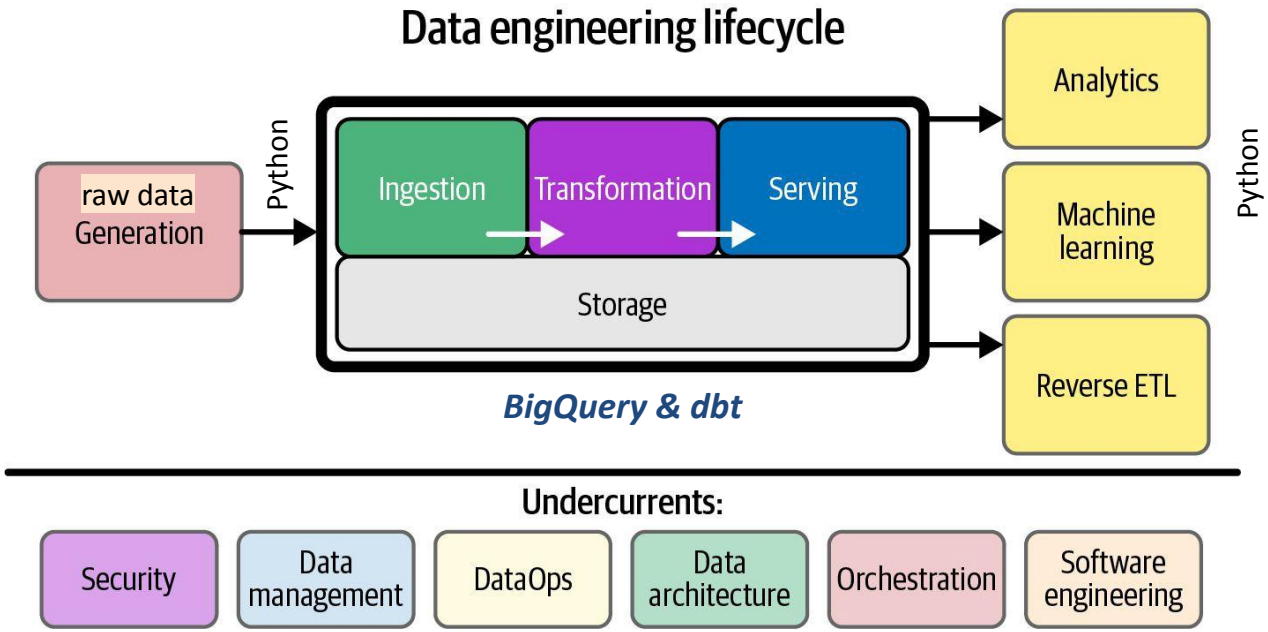
Scalable Data Solution



- End-to-end pipeline: *CSV → BigQuery → dbt → Analysis*
- **Star schema** with **6 dimensions** + **1 fact table**
- **63 data quality tests** across 4 categories
- **7 key analyses** with actionable business insights

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Automated ELT: Technology & Workflow

Tool Choices & Why

- **BigQuery** — serverless, scalable, SQL-native; no infra to manage
- **dbt** — version-controlled SQL transforms, built-in testing, lineage docs
- **Python** + pandas — flexible ingestion with chunked upload for large files
- **Great Expectations** — industry-standard data quality framework
- Git + **GitHub** — full audit trail of all code changes

- 1. **Ingestion** — Python script loads 3 CSV files into BigQuery (ai_adoption_raw)
- 2. **Staging** — dbt views clean and standardize raw tables
- 3. **Marts** — dbt transforms data into Star Schema (6 dims + 1 fact)
- 4. **Testing** — dbt schema tests + Custom SQL + Great Expectations
- 5. **Analysis** — Jupyter Notebook connects via BigQuery client for EDA

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Schema Design

We choose Star Schema over other Schemas because:

- Simple structure : fast analytical queries
- Easy to understand : business stakeholders
- Dimension tables can be reused across multiple fact tables
- Optimized for GROUP BY and aggregation queries in BigQuery

SQL command to Bigquery

1 Ai Survey Fact Table

6 Dimension Table

Context

1. Company : size, industry, country
2. Date : year, quarter, year_quarter
3. AI Tool: tool name, vendor category
4. AI UseCase : use case, category
5. AI AdoptionStage : stage order/label
6. Survey Source : source, methods

To Find out

Capture

1.Revenue growth percent
2.Productivity change percent
3. Task automation rate
4. Jobs displaced / Jobs created
5. Net jobs change
6. Ai roi index

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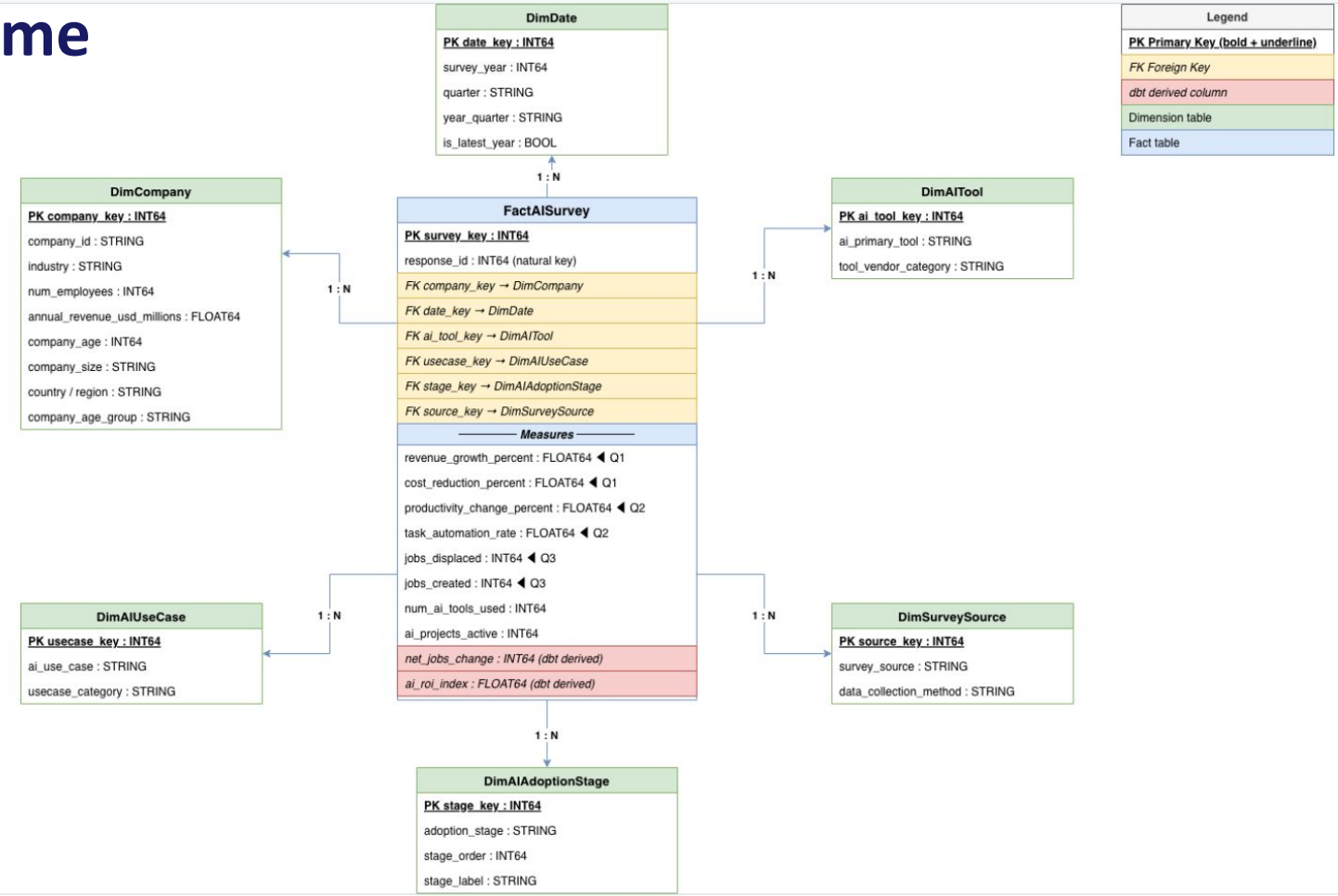
Star Scheme Structure



DATA QUALITY

3-Tier Testing Strategy

63 automated checks
across 150,000 records

100% Pass Rate

TIER 1
dbt Schema Tests

35 tests

- unique + not_null on all PKs
- FK relationships on all fact keys
- accepted_values on categorical fields

TIER 2
Custom SQL Audits

25 tests

- 6 null checks across all tables
- 5 duplicate checks on all PKs
- 6 FK referential integrity checks
- 8 business logic / value range checks

TIER 3
Great Expectations

3 tests

- survey_key not null
- task_automation_rate between 0–100
- jobs_displaced ≥ 0

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Test Results & Business Value

Tier	Tool	Tests	Result
Tier 1	dbt Schema Tests	35	✅ All Pass
Tier 2	Custom SQL Audits	25	✅ All Pass
Tier 3	Great Expectations	3	✅ All Pass
TOTAL	3 Tools Combined	63	✅ 100% Pass

Data Integrity

Zero null PKs, zero duplicates, and 100% FK integrity across all 7 tables.

Pipeline Trust

Executives and analysts can trust every metric and KPI derived from this data.

Scalability

Automated tests run on every dbt build — quality is enforced, not assumed.

Key Takeaway

Our 3-tier quality framework validated 150,000 records with 63 automated checks — achieving a 100% pass rate. This ensures every AI adoption insight and ROI metric in this presentation is built on verified, trustworthy data.

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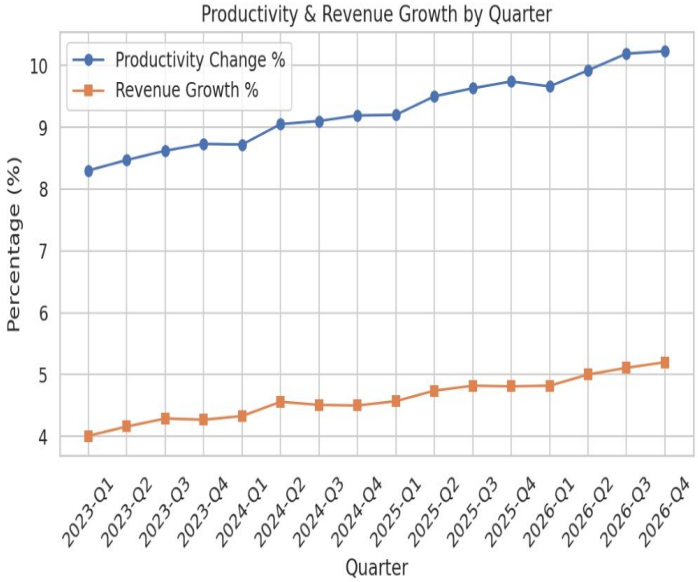
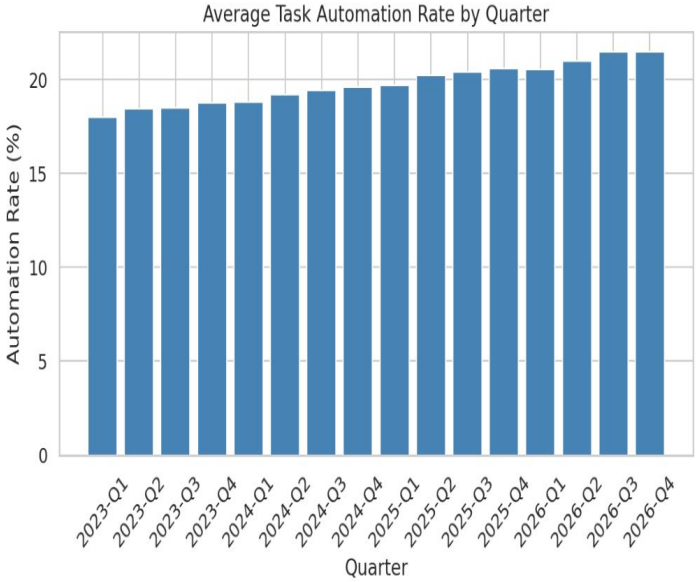
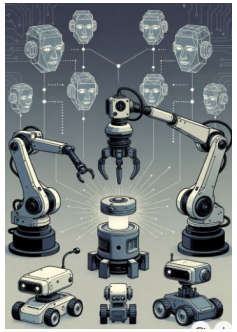
EDA Key Findings

1. Productivity & Revenue Growth and Task Automation Rate

- Productivity and revenue growth trend upward as AI adoption matures across quarters.
- Task automation rate rises steadily from 2023 to 2026, indicating pipeline maturity.







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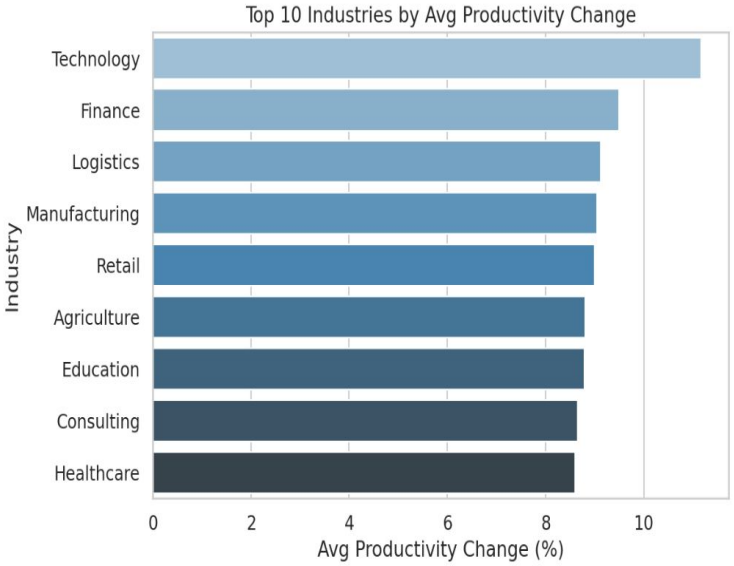
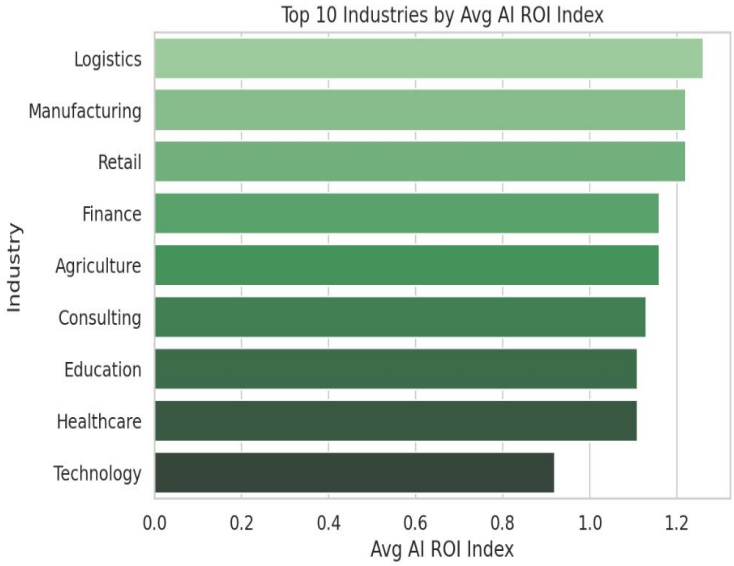
EDA Key Findings

2. Top Industries by Productivity Growth and AI ROI

- Technology, Finance, & Technology show highest avg productivity growth.
- Logistics, Manufacturing & Retail show highest AI-ROI; traditional sectors benefit most.





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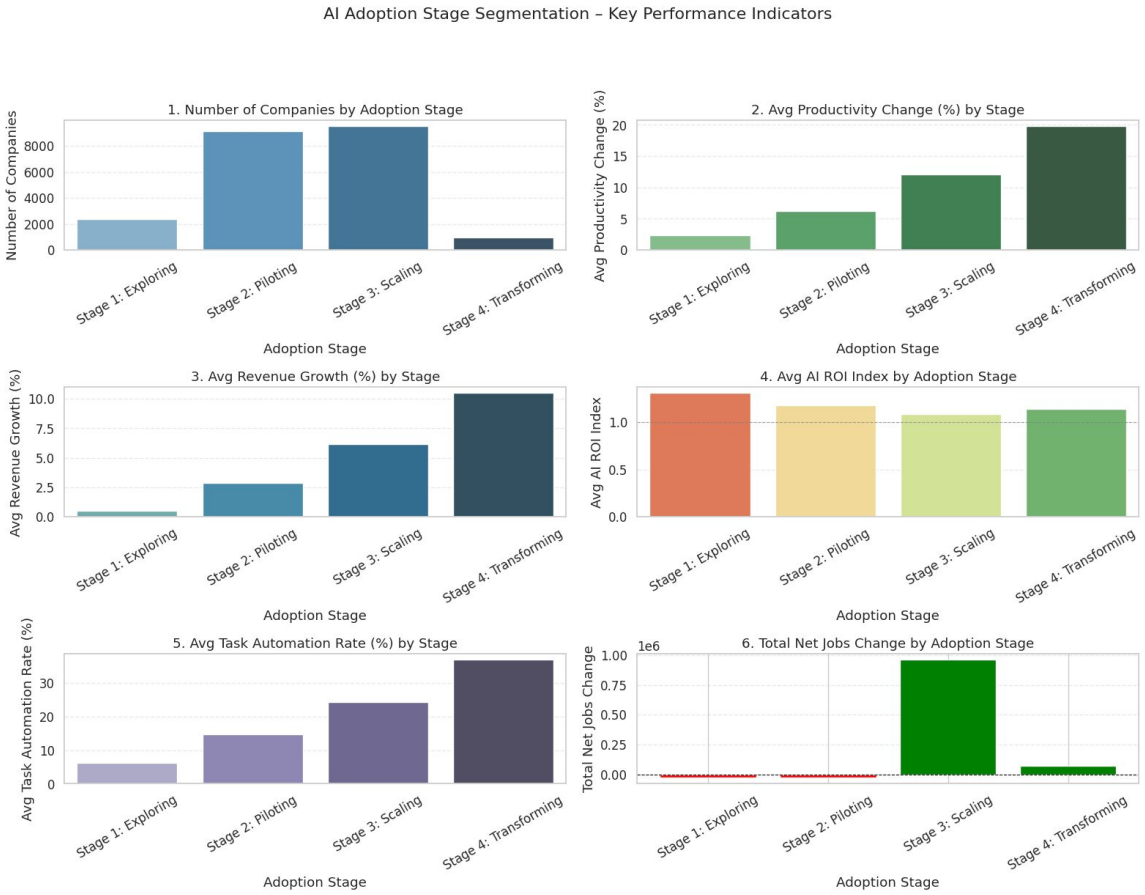
EDA Key Findings

3. Adoption Stage Segmentation

- As AI adoption matures, more companies have transitioned from Exploring to Piloting and Scaling, but most have not gone into Transforming
- Productivity & Revenue growth, Task Automation Rate trend very strongly upward as Exploring -> Piloting -> Scaling -> Transforming
- Total Net Jobs Change: Scaling by far has the most effect (+970k jobs)



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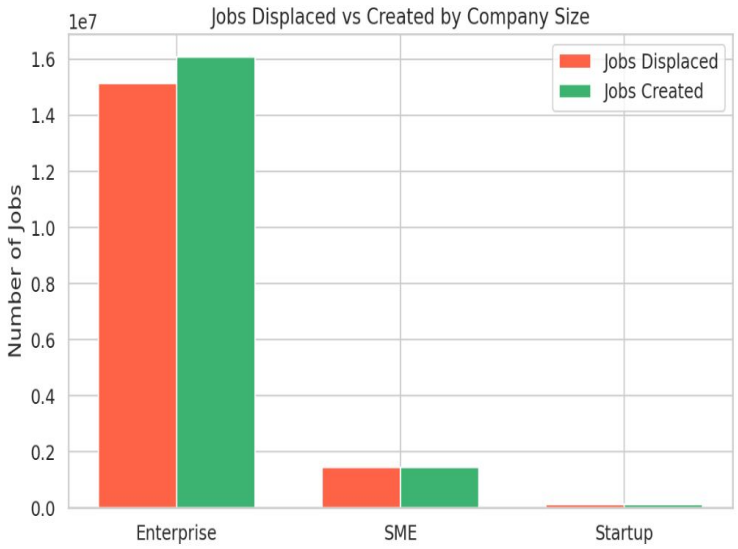
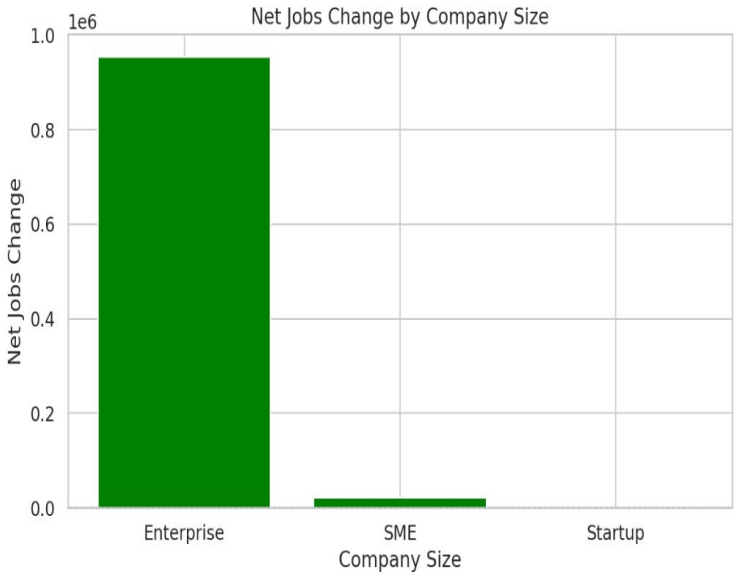
EDA Key Findings

4. Jobs Displaced vs Created and Net Jobs Change by Company Size

- Enterprise by far creates the most jobs but with corresponding jobs displacement as well. Net very positive relative to SMEs & Startups due to sheer size.





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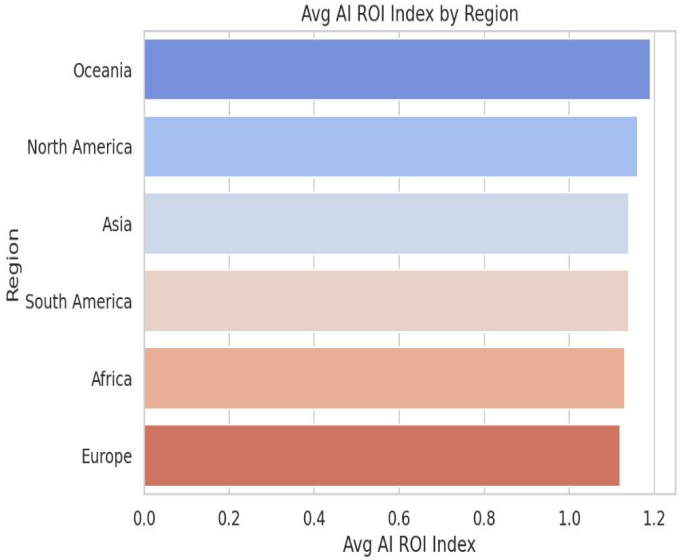
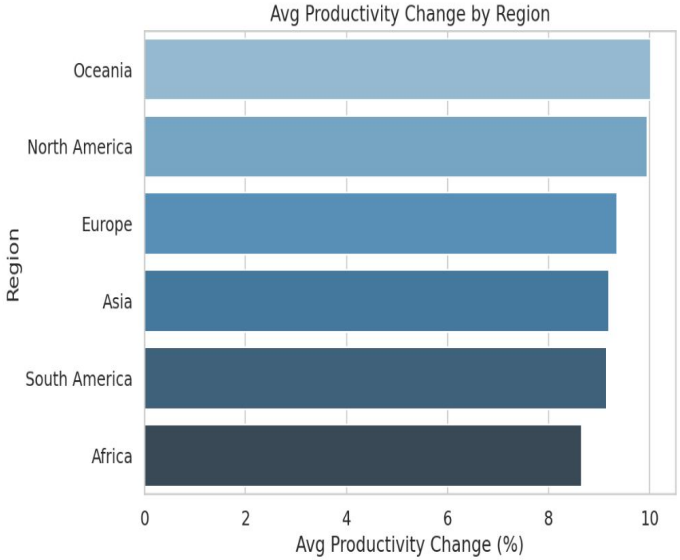
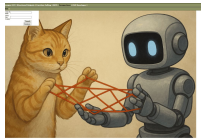
EDA Key Findings

5. Productivity Growth and AI ROI by Region

- Oceania, N. America & Asia / Europe top for both measures. Less developed regions S. America & Africa lag behind -> More developed regions reap more benefits.







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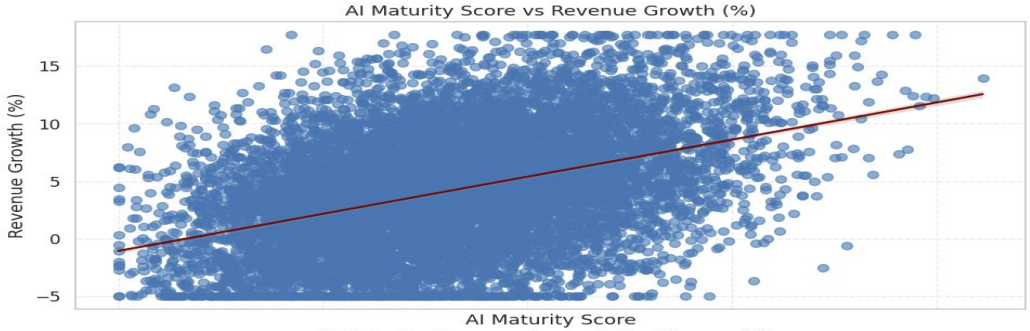
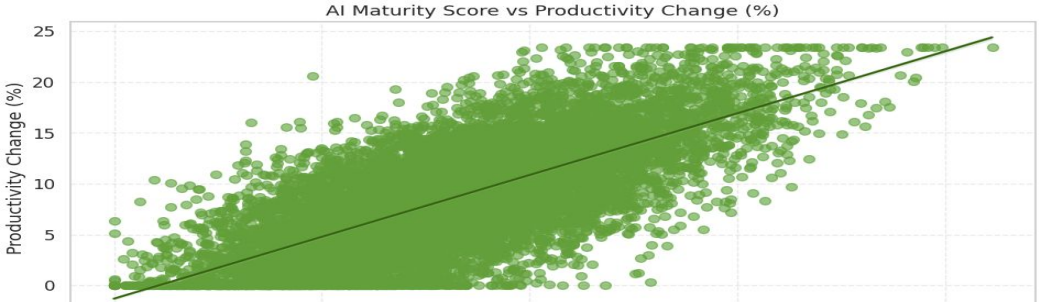
EDA Key Findings

6. Revenue Growth, Productivity Growth, Net Jobs Created by AI Maturity

- All three trend upward with increasing maturity; Productivity highest @ Pearson = 0.728, Revenue growth Pearson =0.41, Net Jobs Created Pearson =0.11
- Productivity has strongest relationship due to immediate nature of measurement -> productivity gain in the specific department as it does the AI project
- Revenue effect is much more dispersed because reporting entire company revenue
- Net Jobs created shows very wide dispersion; at low maturity -> more dispersion, higher maturity -> cluster to zero. As AI matures -> labour-neutral.





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EDA Key Findings

7. Time-Series (One Quarter Lagged) Analysis

- AI Maturity Score dominates for Productivity Growth, which in turn strongly dominates Revenue Growth; even more dominant for Net Jobs Created.
- More active AI projects show negative short-term associations with revenue & productivity -> Quality >> Quantity

1. Revenue Growth (%)

Key coefficient table

Variable	Coefficient	Std. Error	t-stat	p-value	95% CI Lower	95% CI Upper
ai_projects_lag1	-0.0232	0.0036	-6.46	<0.001	-0.0303	-0.0162
ai_maturity_score	-1.1116	0.2191	-5.07	<0.001	-1.5411	-0.6822
ai_adoption_rate	0.0534	0.0016	33.19	<0.001	0.0503	0.0566
ai_budget_percentage	-0.0100	0.0057	-1.74	0.083	-0.0212	0.0013
productivity_change_percent	0.3625	0.0033	109.74	<0.001	0.3560	0.3690

2. Productivity Change (%)

Key coefficient table

Variable	Coefficient	Std. Error	t-stat	p-value	95% CI Lower	95% CI Upper
ai_projects_lag1	-0.0512	0.0029	-17.37	<0.001	-0.0570	-0.0454
ai_maturity_score	24.158	0.1680	143.78	<0.001	23.829	24.488
ai_adoption_rate	0.0639	0.0013	48.18	<0.001	0.0613	0.0665
ai_budget_percentage	-0.0917	0.0047	-19.63	<0.001	-0.1008	-0.0825

3. Net Jobs Change (created – displaced)

Key coefficient table

Variable	Coefficient	Std. Error	t-stat	p-value	95% CI Lower	95% CI Upper
ai_projects_lag1	0.1074	0.0827	1.30	0.194	-0.0546	0.2695
ai_maturity_score	55.848	4.6823	11.93	<0.001	46.670	65.025
ai_adoption_rate	-0.3022	0.0381	-7.93	<0.001	-0.3768	-0.2275
ai_budget_percentage	0.0373	0.1294	0.29	0.773	-0.2163	0.2909

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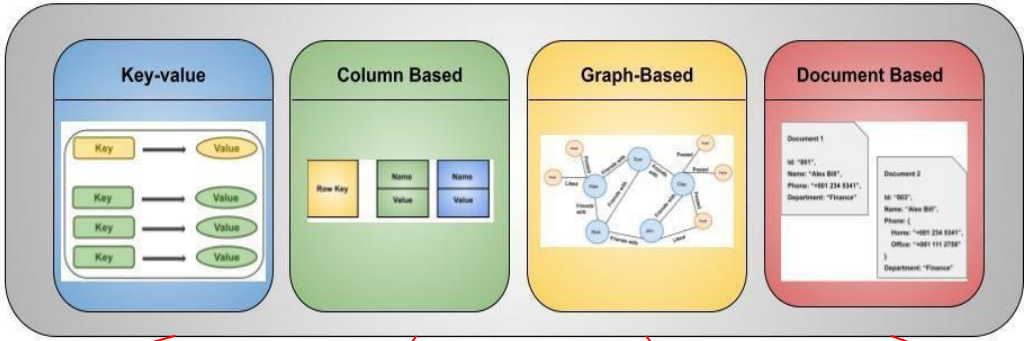
Business Recommendations



1. Prioritise AI Maturity

Companies should start AI projects early and learn fast to maximize maturity. Prioritize Quality over Quantity. Could start with most important value-added stage and expand.

2. Productivity

Aim to increase Productivity of the process / department which is doing the Ai project -> Most immediate and substantial gain. Make Productivity the central focus.

3. Revenue

As company matures in AI, aim to use new capabilities to venture into new areas / revenue streams. E.g. Nvida venturing into Nemoclaw, Robotics, Disney tie-ups etc.

4. Jobs / Workforce Development

Optimize jobs with the most gains from AI, prioritize re-skilling of employees into AI, transfer suitable workers to and create new jobs for new business areas

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Risk & Mitigation

RISK	MITIGATION
CSV file is 40MB — loading all at once could crash Python	Streamed data in chunks of 10,000 rows using pandas
BigQuery connection could fail during ingestion	Added try/except error handling — failures are logged and skipped automatically
Raw data had inconsistent adoption stage values ('none', 'pilot')	Added a CASE mapping in dbt to standardise all values before loading
Data could have null values or duplicate records	Ran 63 automated tests covering null checks, duplicates, and referential integrity
Service account key could be accidentally pushed to GitHub	Stored in .env file and added to .gitignore — never committed to Git
Dataset is simulated, not real company data	All insights are clearly labelled as directional estimates, not definitive conclusions

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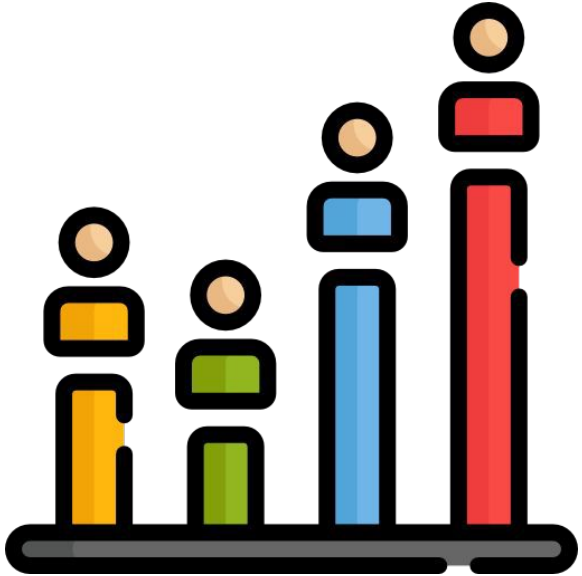
Thank You

Questions & Discussion

GitHub: https://github.com/sheenZ2026/ai-adoption-pipeline

Dataset: Kaggle — Global AI Adoption & Workforce Impact
(https://www.kaggle.com/datasets/mohankrishnathalla/global-ai-adoption-and-workforce-impact-dataset?resource=download)

Stack: BigQuery · dbt · Python · Great Expectations · Jupyter



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